

Note 2.2 Limit of a Function and Limit Laws

Limits of Function Values

- **Definition:**

Suppose $f(x)$ is defined on neighborhood of x_0 except possibly x_0 , as x approaches x_0 from either side, $f(x)$ gets sufficiently close to L . Then L is the limit of $f(x)$ as x approaches x_0 .

Denoted by

$$\lim_{x \rightarrow x_0} f(x) = L \quad \text{or} \quad f(x) \rightarrow L (x \rightarrow x_0)$$

- **The cases of no limitation:**

- Jump: $U(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$ when $x \rightarrow 0$
- Infinity: $g(x) = \frac{1}{x}$ when $x \rightarrow 0$
- oscillate: $f(x) = \sin \frac{1}{x}$ when $x \rightarrow 0$

Limit Laws

- **Limit Laws:**

If L, M, C, k are real numbers, and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M$$

then

$$\lim_{x \rightarrow c} (f(x) \pm g(x)) = L \pm M$$

$$\lim_{x \rightarrow c} kf(x) = kL$$

$$\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$$

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, M \neq 0$$

$$\lim_{x \rightarrow c} [f(x)]^n = L^n, n > 0$$

$$\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L}, n > 0 \text{ (If } n \text{ is even, we assume that } \lim_{x \rightarrow c} f(x) = L)$$

- **Limits of Polynomials:**

If $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$, then

$$\lim_{x \rightarrow c} P(x) = P(c) = a_n c^n + a_{n-1} c^{n-1} + \dots + a_0$$

- **Limits of [Rational Functions](#):**

If $P(x)$ and $Q(x)$ are polynomials and $Q(c) \neq 0$, then

$$\lim_{x \rightarrow c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}$$

Eliminating Zero Denominators Algebraically

- **For rational function:**

$$\lim_{x \rightarrow c} \frac{P_n(x)}{Q_n(x)} = \lim_{x \rightarrow c} \frac{(x - c)P_{n-1}(x)}{(x - c)Q_{n-1}(x)} = \lim_{x \rightarrow c} \frac{P_{n-1}(x)}{Q_{n-1}(x)}, P_n(c) = Q_n(c) = 0$$

- **For irrational function:**

Rationalize the numerators or denominators first by formulas:

$$(\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b}) = a - b$$

$$(\sqrt[3]{a} - \sqrt[3]{b})(\sqrt[3]{a^2} + \sqrt[3]{ab} + \sqrt[3]{b^2}) = a - b$$

$$(\sqrt[n]{a} - \sqrt[n]{b}) = (\sqrt[n]{a^{n-1}} + \sqrt[n]{a^{n-2}b} + \dots + \sqrt[n]{b^{n-1}}) = a - b$$

The Sandwich Theorem

- **Definition:**

If $g(x) \leq f(x) \leq h(x)$ when $x \rightarrow c$, and $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L$, then

$$\lim_{x \rightarrow c} f(x) = L$$

- **Theorem 5:**

If $f(x) \leq g(x)$ for all x in some opening interval containing c , except possibly at $x = c$ itself, and the limits of f and g both exist as x approaches c , then

$$\lim_{x \rightarrow c} f(x) \leq \lim_{x \rightarrow c} g(x)$$

- Corollary: If $f(x) \geq 0$, and $\lim_{x \rightarrow c} f(x)$ exists, then $\lim_{x \rightarrow c} f(x) \geq 0$